

Program: Diploma in Fire Technology and Safety	
Course Code: 4462	Course Title: Environmental Safety
Semester: 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To provide students with a foundational understanding of environmental safety concepts, focusing on health, safety, and environmental (HS&E) practices.
- To develop skills in safety planning, risk management, and applying leadership principles for accident prevention.
- To educate students on how to assess hazards, control risks, and create a safe working environment.

Course Pre-requisites:

Topic	Course Code	CourseName	Semester
Principles of Safety Management		Principles of Safety Management	2

CourseOutcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	CognitiveLevel
CO1	Understand the basics of health, safety, and environmental issuesand how to manage them.	13	Understanding
CO2	Plan and develop safety strategies and implement safety measures in the workplace	15	Applying
CO3	Apply safety management systems, leadership principles, and communicate safety effectively.	15	Applying

CO4	Assess and control safety and health hazards, preventing accidents and promoting workplace safety.	15	Applying
	Series Test	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	1			
CO2	3	3	3	2			
CO3	3	2	3	2			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the basics of health, safety, and environmental issues and how to manage them.		
M1.01	Introduction to environmental safety, HS&E representatives, pesticides.	3	Understanding
M1.02	Steps in safety planning and developing safety network.	6	Applying
M1.03	Creating and monitoring workplace safety plans.	4	Applying

Contents:

Introduction: Overview of environmental safety, HS&E representatives, and the role of environmental safety in the workplace.
Health, Safety, and Environment for “Grassroots” Members: Safety and environmental practices for workers at all levels.
Health, Safety, and Environment for Worker HS&E Representatives: Responsibilities and roles of safety representatives.
Pesticides and Health: Understanding the impact of pesticides on health and the environment.
Health, Safety, and Environment Fact Sheets: Key documents outlining safety and environmental guidelines.

Planning for Safety: Steps involved in safety planning, from identifying hazards to developing a safety plan. Building an Informed Family Safety Network: Involvement of families and communities in safety planning. Monitoring Progress: Keeping track of safety progress in the workplace.			
CO2	Plan and develop safety strategies and implement safety measures in the workplace		
M2 No	Description	Duration (Hours)	Cognitive Level
M2.01	Overview of safety management systems and hazard management.	5	Understanding
M2.02	Role of communication and safety audits in organizations.	5	Applying
M2.03	Principles of safety leadership and roles of safety directors, managers, and team leaders.	5	Applying
	SeriesTestI	1	
Contents: Safety Management Systems: Understanding the structure and components of safety management in organizations. Organization and Communication: Effective communication strategies for safety at the workplace. Hazard Management: Identifying, assessing, and managing safety hazards in the environment. Safety Auditing: Importance of safety audits and tools used in safety evaluations. Directing for Safety: The role of directors, managers, and team leaders in promoting safety within the organization. The Principles of Safety Leadership: Key aspects of leadership that ensure a safe work environment, including safety directives from plant directors, project managers, and maintenance managers.			
CO3	Apply safety management systems, leadership principles, and communicate safety effectively..		
M3.01	Safety foundations, organizing safety, risk assessment.	5	Applying
M3.02	Control of physical, mechanical, fire, and electrical hazards.	5	Applying

M3.03	Hazards: manual handling, psychological health, and fire control.	5	Applying
Contents: Health and Safety Foundations: Basics of maintaining health and safety within organizations and the community. Risk Assessment: Analyzing workplace hazards and assessing risks. Manual and Mechanical Handling Hazards and Control: Identification and control measures for handling equipment and machinery safely. Physical and Psychological Health Hazards: Addressing potential physical and psychological risks in the workplace. Fire Hazards and Control: Managing fire risks through preventive measures, training, and control systems. Electrical Hazards and Control: Ensuring safety from electrical accidents and applying preventive measures. Principles of Control: Methods for controlling hazards and minimizing risks in the workplace.			
CO4	Assess and control safety and health hazards, preventing accidents and promoting workplace safety.		
M4.01	Accident prevention, safety programs, audits.	5	Understanding
M4.02	Investigating accidents and job safety analysis.	5	Applying
M4.03	Safety training and hazard prevention programs.	5	Applying
	SeriesTest II	1	
Contents: Safety and Health Programs: Designing and implementing workplace safety and health programs. Accident and Incident Investigation: Procedures for investigating accidents and incidents in the workplace. Job Safety/Hazard Analysis: Identifying job-specific hazards and implementing safe work procedures. Safety and Health Audits: Regular audits and evaluations of safety standards in the workplace. Safety and Health Training: Developing safety training programs for employees at all levels. Health Hazard Prevention: Techniques for preventing health-related hazards in the workplace. Safety Hazards Prevention: Identifying common safety hazards and methods to itigate			

Text/Reference:

T/R	BookTitle/Author
T1	Environmental Safety – Charles D. Reese
T2	Health, Safety, and Environment Handbook – J.G.M.S.T. Smith
R1	Occupational Safety and Health – David L. Goetsch
R2	Introduction to Safety Management – J.J.K. Poel
R3	Safety and Health at Work – John R. Ridley